

Harkirat Singh

AI Financial Analyst Copilot – Project Report

Executive Summary

The AI Financial Analyst Copilot is a publicly deployed AI-powered stock analysis application built using Streamlit, Yahoo Finance, and OpenAI. Users can analyze stocks, compare multiple tickers, view KPIs, and receive natural-language analyst summaries. The project evolved from a local prototype into a production-ready cloud application deployed on Streamlit Cloud.

What the App Can Do

- Pull live stock market data using Yahoo Finance.
- Analyze multiple tickers simultaneously.
- Display KPI metrics including current price, returns, volatility, and trend signals.
- Compare companies side-by-side.
- Generate AI-powered analyst summaries using OpenAI.
- Allow natural language questions from users.
- Operate through a publicly accessible web interface.

Business Impact

This application demonstrates how AI can augment financial analysis by converting quantitative market data into understandable narratives. It reduces the barrier to interpreting financial indicators and illustrates how LLMs can support decision-making workflows.

Architecture

User → Streamlit Frontend → Yahoo Finance API → KPI Calculations → OpenAI API → AI Summary → Streamlit Cloud Deployment

Major Challenges and Debugging Journey

- ModuleNotFoundError: yfinance not installed.
- Streamlit not installed in the virtual environment.
- pyarrow wheel installation failures.
- GitHub authentication issues (403 errors).
- Streamlit deployment dependency failures.
- requirements.txt generated from pip freeze caused conflicts.
- Plotly installation issues in Streamlit Cloud.
- Streamlit cache serving stale deployments.
- Python version inconsistencies.
- Hidden Plotly build_chart function causing NameError.

- TOML syntax errors when configuring Streamlit Secrets.
- Missing OpenAI API key in production.

Key Engineering Lessons

- Local environments differ from production environments.
- Dependency management is critical.
- Cloud deployments require secrets management.
- Debugging and persistence are core engineering skills.
- Shipping software involves integration, deployment, and operations.

Future Enhancements

- Add sentiment analysis from financial news.
- Support earnings call summaries.
- Add portfolio tracking.
- Implement watchlists and alerts.
- Add authentication and user accounts.
- Include advanced forecasting models.
- Migrate deployment to AWS.

Reflection

Although the app initially appeared simple, the deployment process revealed the realities of software engineering. The project required persistence, iterative debugging, and systems thinking. The most valuable outcome was not merely the application itself, but gaining confidence in the ability to transform an idea into a deployed product.

Conclusion

The AI Financial Analyst Copilot represents a significant milestone: the successful design, development, debugging, and deployment of a real-world AI application accessible to end users.